CUHK-Shenzhen
Course Outline

1. Course Identity

A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MFM5220
Course title (English)	Algorithmic Investment
Course title (Chinese)	演算法投資
Units	3
Description (English)	This course introduces mathematical models and practice for algorithmic and high-frequency trading. The topics includes basics of stochastic control and optimal stopping, optimal execution, market making, pairs trading, and statistical arbitrage, etc.
Description (Chinese)	本課程主要介紹演算法與高頻交易的數學模型与实践。內容包括隨機控制與最優停損問題、最優執行、做市策略、配對交易、統計套利等。

B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in SZ maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	FTEC5530
Course title (English)	Quantitative and Algorithmic Trading
Course title (Chinese)	量化與演算法交易
Units	3
Description (English)	The industry landscape of investment, trading, and risk management has been revolutionized by computing technologies, data science, and financial engineering. To progress in tandem with the changes in the industry, the topics covered in this course leverage on the recent developments in portfolio theory, empirical finance, and quantitative and algorithmic trading. In addition to mathematical modeling, an important part of this course is the practical aspect: computational implementations with statistical tests. Given that implementation and test procedures are involved, this quantitative finance course is algorithmic and hands-on in nature.
Description (Chinese)	產業投資、交易及風險管理的格局已因計算科技、資料科學與金融工程而徹底革新。為了與產業變化同步，本課程所涵蓋的主題，將利用近期投資組合理論、實證金融，以及量化與演算法交易的最新發展。除了數學建模之外，本課程的一個重要部分是實務面：透過統計檢驗進行電腦實作。鑒於課程包含實作與測試流程，這門量化金融課程屬於演算法性質，且強調動手操作。

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1. **Prerequisites**

Stochastic processes, differential equations.

A. Co-requisites

None.

2. Learning Outcomes

Students will be able to:
Understand the basics of financial products.
Understand the functioning of financial markets.
Understand the empirical financial data characteristics.
Understand major mathematical models for optimal execution.
Understand major mathematical models for market making.
Understand major mathematical models for pairs trading and statistical arbitrage.

3. Course syllabus

Introduction.
Preliminaries on financial products, markets, and trading.
Stochastic control and optimal stopping.
Optimal execution.
Market making.
Pairs trading and statistical arbitrage.
Reinforcement learning.

4. **Assessment Scheme**

Component/ method	**% weight**
Assignments	30
Midterm	20
Final project (individual)	50

5. Descriptor

Grade	Overall course
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

6. **Declaration on the use of AI tools.**

Are students permitted to use AI tools (in particular GPT and generative AI tools) in the learning process and their assignment/course project? Please select one of the following approaches that will be adopted in this course.

☐ Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course.

■ Use only with explicit acknowledgement: Students are allowed or expected to use AI tools in their learning process and assignment/project reports. However, students must explicitly cite or acknowledge these resources and explain how they were used in their submitted reports. The course instructor will provide specific examples. Failure to follow this rule will be considered academic dishonesty.

☐ Free use of AI tools with no acknowledge: This course requires frequent use of AI tools, and students can use these tools throughout the learning process without the need to explicitly cite or acknowledge them. The instructor may specify which AI tools are allowed and set proper regulations if needed. Students who do not follow the regularations will be considered academic dishonesty.

Remark: (The instructor should add additional details here about the use of AI tools in this course)

7. **Feedback for evaluation**

A course questionnaire (CTE) was answered by each student after completing all teaching activities.

Reflections from teachers of more advanced courses.

Feedback from informal conversations over in the instructor's office and wechat/e-mail correspondence with students after class.

8. Reading

A. Required

Cartea, Álvaro, Sebastian Jaimungal, and José Penalva. Algorithmic and high-frequency trading. Cambridge University Press, 2015.

B. Recommended

Chan, Ernie. Algorithmic trading: winning strategies and their rationale. John Wiley & Sons, 2013.
Hilpisch, Yves. Python for Algorithmic Trading. O'Reilly Media, 2020.

9. Course components

Activity	Hours/week
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Introduction
2	Preliminaries on financial products, markets, and trading
3	Stochastic control and optimal stopping (I)
4	Stochastic control and optimal stopping (II)
5	Stochastic control and optimal stopping (III)
6	Optimal execution (I)
7	Optimal execution (II)
8	Mid-Term Exam
9	Market making (I)
10	Market making (II)
11	Pairs trading and statistical arbitrage (I)
12	Pairs trading and statistical arbitrage (II)
13	Advanced topics on algorithmic trading (I)
14	Advanced topics on algorithmic trading (II)
15	Final project